

A common root vector realizes every selected state through the generators

MathlibAnnex.CStarAlgebra.CAR.exists_targetDefectVectorFamily

theorem

Builds a compatible family of unit vectors from a surviving residual space and identifies their CAR vector states.

Statement

For the supplied shell family and nonzero irreducible representation ρ described below, there exist a unit vector $\zeta \in K$ and unit vectors $\eta_j \in K$, one for each $j \in I$, with the following properties for every n and $a \in A$:

$$\sigma(f_n)\zeta = \zeta, \quad \langle \zeta, \sigma(a)\zeta \rangle = \omega(a).$$

For every $j \in I$ they satisfy

$$\sigma(f_{j,n})\eta_j = \eta_j, \quad V_j\eta_j = \zeta, \quad \langle \eta_j, \sigma(a)\eta_j \rangle = \varphi_j(a).$$

Here $\sigma = \rho \circ \iota$, and inner products are linear in the second argument.

Assumptions

Let A be the completed CAR algebra, obtained by completing the matrix stages $M_{2^n}(\mathbb{C})$ with embeddings $a \mapsto a \otimes I_2$. Let f_n be the image of the first diagonal matrix unit. Thus $f_0 = 1$ and the f_n form a decreasing sequence of projections. The root state ω has value a_{00} on a matrix a in any stage.

Choose a pure state φ_j in each unitary-equivalence class $j \in I$ of pure-state GNS representations, with $\varphi_o = \omega$ at the root index o . Denote its complete complex GNS space, unital star representation and unit cyclic vector by (H_j, π_j, ξ_j) . Set $\mathcal{H} = \bigoplus_{j \in I} H_j$ and $\pi = \bigoplus_{j \in I} \pi_j$. The index set need not be countable.

A supplied shell family consists of unital complex star automorphisms α_j with $\varphi_j \circ \alpha_j = \omega$ and elements $w_{j,n} \in A$. Put $f_{j,n} = \alpha_j(f_n)$. Their support identities are $w_{j,n}^* w_{j,n} = f_{j,n} - f_{j,n+1}$ and $w_{j,n} w_{j,n}^* = f_n - f_{n+1}$. At the root, α_o is the identity and $w_{o,n} = f_n - f_{n+1}$.

Let L_j be unitaries on $\mathcal{H}$ satisfying $L_j \pi(f_{j,n} - f_{j,n+1}) = \pi(w_{j,n})$ for every j, n . Let $T \subseteq B(\mathcal{H})$ be the norm-closed unital star algebra generated by $\pi(A)$ and all L_j . Write $\iota : A \rightarrow T$ for $\iota(a) = \pi(a)$ and $g_j \in T$ for the element represented by L_j . For a unital complex star representation $\rho : T \rightarrow B(K)$ on a complete complex Hilbert space, put $\sigma = \rho \circ \iota$ and $V_j = \rho(g_j)$. Each V_j is unitary because ρ preserves the unit, products and adjoints. Write $F_j = \bigcap_{n \geq 0} \text{ran } \sigma(f_{j,n})$ for the common fixed space of the j -th transported flag. The representation ρ is nonzero and irreducible, in the sense that its only closed reducing subspaces are $\{0\}$ and K . No root normalization $L_o = I$ and no prescribed action of L_j on the atomic cyclic vectors is required.

Conclusion

All chosen vector states share one root vector: the family can be defined by $\eta_j = V_j^* \zeta$. These pointed state realizations supply the vector data for the cyclic-sum capture theorem. The present result does not assert that an individual η_j is cyclic for all of K , or that the limiting fixed spaces are one-dimensional.

The common root vector ζ and the member η_o indexed by the root class have the same root state, but their equality is not imposed here. The additional normalization $L_o = I$, used in that cyclic-sum capture theorem, would give $V_o = I$ and hence $\eta_o = \zeta$. No orthogonality or surjectivity conclusion is included in this declaration.

Proof route

Choose a unit vector in one nonzero limiting fixed space and apply its represented generator to obtain ζ in the root fixed space. The CAR norm-compression estimates identify both vector states. For every other index, pull this same ζ back by the corresponding unitary. The fixed-space equivalence in shell reconstruction places the pulled-back vector in the correct fixed space, and compression again identifies its state.

▼ Proof steps

1. The nonvanishing theorem gives i_0 and $0 \neq x \in F_{i_0}$. Normalize x to $\eta^{(0)} = x/\|x\|$ and put $\zeta = V_{i_0} \eta^{(0)}$. Unitarity preserves norm one. Reconstruction maps F_{i_0} onto the root common range, so all $\sigma(f_n)$ fix ζ .
2. For every $a \in A$, the source estimates are $\|f_n a f_n - \omega(a) f_n\| \rightarrow 0$ and $\|f_{j,n} a f_{j,n} - \varphi_j(a) f_{j,n}\| \rightarrow 0$. Apply the contractive representation σ and take a matrix coefficient at a unit vector fixed by the corresponding projections. Self-adjointness and fixedness make that coefficient equal to the constant difference between its vector state and the indicated source state. Norm convergence forces the difference to be zero. In particular this identifies the state of ζ as ω .
3. Define $\eta_j = V_j^* \zeta$ for each j . The same reconstruction equivalence, used in the reverse direction, gives $\eta_j \in F_j$. Unitarity gives $\|\eta_j\| = 1$ and $V_j \eta_j = \zeta$; membership in the ranges of projections gives their fixedness.
4. Apply the transported compression estimate to this unit fixed vector. It yields $\langle \eta_j, \sigma(a) \eta_j \rangle = \varphi_j(a)$ for every $a \in A$ and every j . The construction uses one root vector throughout rather than separately choosing unrelated vector-state realizations.

Main citations

- Exact theorem and proof
- The selected states and shell data
- A nonzero initial fixed space exists
- Normalization and the two identified vector states
- The initial transported pair in an irreducible representation
- Transport between the two limiting fixed spaces
- Norm compression toward each selected state

- The root compression estimate as the identity-automorphism case
- Use in the surjective cyclic-sum capture theorem
- The root state bundled with its purity proof

Lean source signature (exact)

```
theorem exists_targetDefectVectorFamily
  (family : RepresentativeShellFamily)
  (L : MathlibAnnex.CStarAlgebra.PureState.GNSClass Limit →
    MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert completedRootPureState
→L[ $\mathbb{C}$ ]
    MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert completedRootPureState)
  (hLunit :  $\forall i, L i \in \text{unitary}$ 
    (MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert completedRootPureState
→L[ $\mathbb{C}$ ]
    MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert completedRootPureState))
  (hLsource :  $\forall i n, (L i).comp (selectedAtomicRepresentation
    (transportedFlag family i n - transportedFlag family i (n + 1))) =
    representedShellLink family i n)$ 
  {K : Type v} [NormedAddCommGroup K] [InnerProductSpace  $\mathbb{C}$  K]
  [CompleteSpace K] (rho : Representation (AtomicTarget L) K)
  (hrho : rho.IsIrreducible) :
   $\exists (\eta_o : K) (\eta : \text{MathlibAnnex.CStarAlgebra.PureState.GNSClass Limit} \rightarrow K),$ 
   $\|\eta_o\| = 1 \wedge$ 
   $(\forall n, (\text{restrictedRepresentation } L \text{ rho}) (\text{rootFlag } n) \eta_o = \eta_o) \wedge$ 
   $\text{Representation.vectorFunctional } (\text{restrictedRepresentation } L \text{ rho}) \eta_o =$ 
   $\text{rootState} \wedge$ 
   $\forall i,$ 
   $\|\eta i\| = 1 \wedge$ 
   $(\forall n, (\text{restrictedRepresentation } L \text{ rho})$ 
   $(\text{transportedFlag family } i n) (\eta i) = \eta i) \wedge$ 
   $(\text{Unitary.linearIsometryEquiv}$ 
   $(\text{representedGeneratorUnitary } L \text{ hLunit rho } i) : K \simeq_{\mathbb{C}} K) (\eta i) =$ 
   $\eta_o \wedge$ 
   $\text{Representation.vectorFunctional } (\text{restrictedRepresentation } L \text{ rho}) (\eta i) =$ 
   $(\text{MathlibAnnex.CStarAlgebra.PureState.representative completedRootPureState } i).1$ 
```

In the source	Mathematical meaning
family : RepresentativeShellFamily	The given CAR shell family $(\alpha_j, w_{j,n})$, with $f_{j,n} = \alpha_j(f_n)$ and the initial and final support identities in the assumptions.
L; hLunit : $\forall i, L i \in \text{unitary}$ (...)	The given unitary family L_j on the selected atomic Hilbert space $\mathcal{H} = \bigoplus_j H_j$.

In the source	Mathematical meaning
<code>hLsource : ∀ i n, (L i).comp (...) = representedShellLink family i n</code>	For every j, n, the given shell relation is $L_j \pi(f_{j,n} - f_{j,n+1}) = \pi(w_{j,n})$, where $\pi = \bigoplus_j \pi_j$. Neither $L_o = I_{\mathcal{H}}$ nor a prescribed action $L_j(e_j \xi_j) = e_o \xi_o$ is an input hypothesis here; the root conditions inside the supplied shell family are unchanged.
<code>rho : Representation (AtomicTarget L) K; hrho : rho.IsIrreducible</code>	The nonzero irreducible unital star representation $\rho : T \rightarrow B(K)$ of $T = C^*(\pi(A), \{L_j\})$, on a complete complex Hilbert space K in an arbitrary universe. Irreducibility means no proper nonzero closed reducing subspace.
<code>restrictedRepresentation L rho</code>	The source restriction $\sigma = \rho \circ \iota : A \rightarrow B(K)$, where $\iota(a) = \pi(a) \in T$. Nonzero irreducibility of ρ is not an assertion of irreducibility of this restriction.
<code>∃ (eta_o : K) (eta : GNSClass Limit → K)</code>	One common root vector $\zeta \in K$ and one family $(\eta_j)_{j \in I}$ satisfy every following condition. <code>eta_o</code> means ζ, not the family evaluated at root class o.
<code> eta_o = 1</code>	The common vector has $\ \zeta\ = 1$.
<code>∀ n, (restrictedRepresentation L rho) (rootFlag n) eta_o = eta_o</code>	For every n, $\sigma(f_n)\zeta = \zeta$.
<code>Representation.vectorFunctional (restrictedRepresentation L rho) eta_o = rootState</code>	The entire functional is the root state: $\langle \zeta, \sigma(a)\zeta \rangle = \omega(a)$ for every $a \in A$.
<code>∀ i, eta i = 1</code>	Each member η_j of the same family has norm one.
<code>∀ n, (restrictedRepresentation L rho) (transportedFlag family i n) (eta i) = eta i</code>	For that same j and every n, $\sigma(f_{j,n})\eta_j = \eta_j$.
<code>(Unitary.linearIsometryEquiv (representedGeneratorUnitary L hLunit rho i) : K ≈_i[C] K) (eta i) = eta_o</code>	The represented unitary $V_j = \rho(g_j)$ sends η_j to the very same ζ for every class j. The wrapper changes no operator values.
<code>Representation.vectorFunctional (restrictedRepresentation L rho) (eta i) = (PureState.representative completedRootPureState i).1</code>	For each j, the functional $a \mapsto \langle \eta_j, \sigma(a)\eta_j \rangle$ equals the selected representative state φ_j. Inner products are linear in their second argument. This states no orthogonality or cyclicity in all of K.

Further source notes: Here `Limit` is A , `completedRootPureState` is the root state ω bundled with its purity proof, and `SelectedAtomicHilbert completedRootPureState` is the atomic Hilbert space $\mathcal{H}$. `eta_o` is the common root vector ζ , while `eta i` is the vector η_j at the source index i corresponding to the mathematical index j ; `eta_o` does not mean `eta` evaluated at the root index.

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▼ **Exact content identity**

Declaration: `MathlibAnnex.CStarAlgebra.CAR.exists_targetDefectVectorFamily`

Accepted content SHA-256: `103e733200a05d065998d0e008adc8f7f0665823b86c9b1c9735671c51a2b9ec`

Accepted source guide SHA-256: `6f1a859e74f1e391b46ee05710558e65545a9ef21e20d6ea3e7e27757f1762a6`

Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`